Machine Learning & High Frequency Time Series

Predicting short-term regime shifts via Hidden Markov Models

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Problem Formulation: High-Frequency FX

The Challenge

Data: Tick-by-tick FX prices at high frequencies

Issue: Bid-ask bounce and async trading induce spurious negative autocorrelation

Goal: Separate latent efficient price from microstructure noise via HMMs

Mathematical Framework

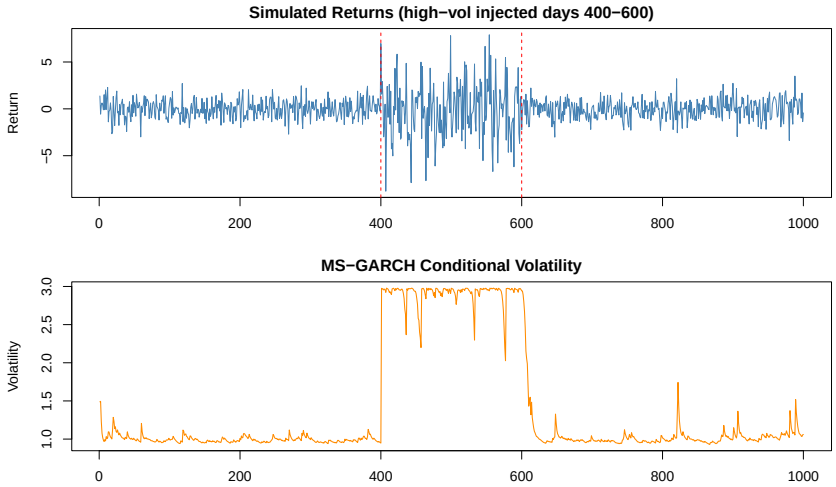
$$P_\tau = P_\tau^* + \eta_\tau$$
$$r_t = \mu_k + \phi_k r_{t-1} + \varepsilon_t$$
$$p_{ij} = \mathbb{P}(S_t{=}j \mid S_{t-1}{=}i)$$

P_τ^*: Latent efficient price
η_τ, ε_t: Noise
r_t: Log return at time t
μ_k: Mean return in regime k
ϕ_k: Persistence in regime k
S_t: Unobserved state (mom/noise)
p_{ij}: Transition prob. from state i to j

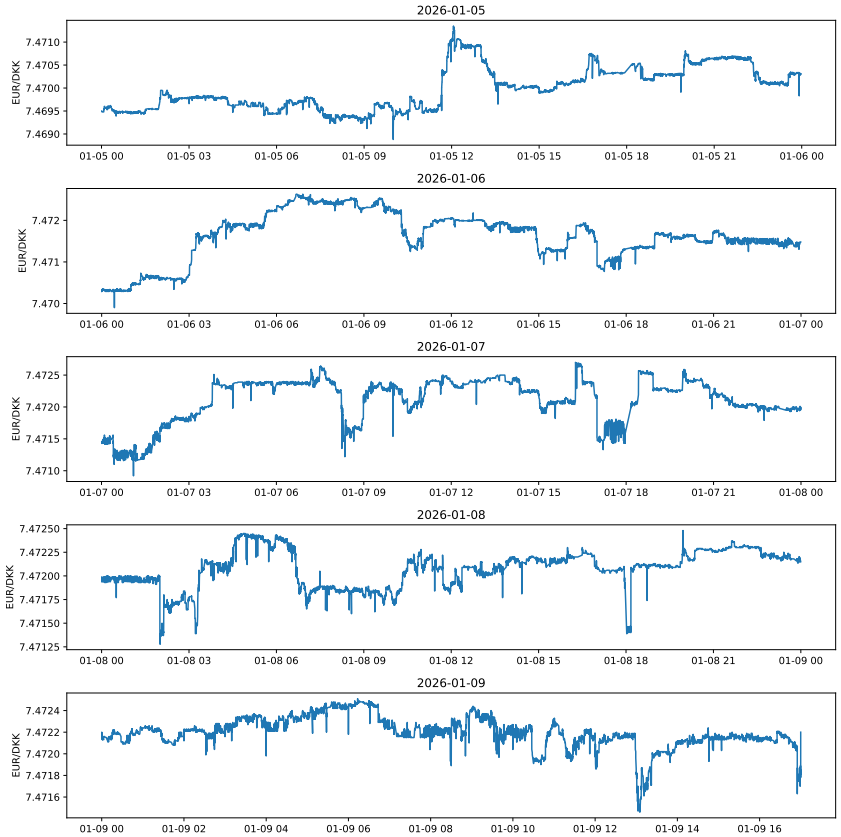
By modeling unobserved states (S_t), the HMM dynamically adapts to time-varying dependence and shifting market liquidity.

Theoretical Case: Microstructure

- Stylized volatility clustering from simulated MS-GARCH
- Empirical EUR/DKK exchange rate dynamics

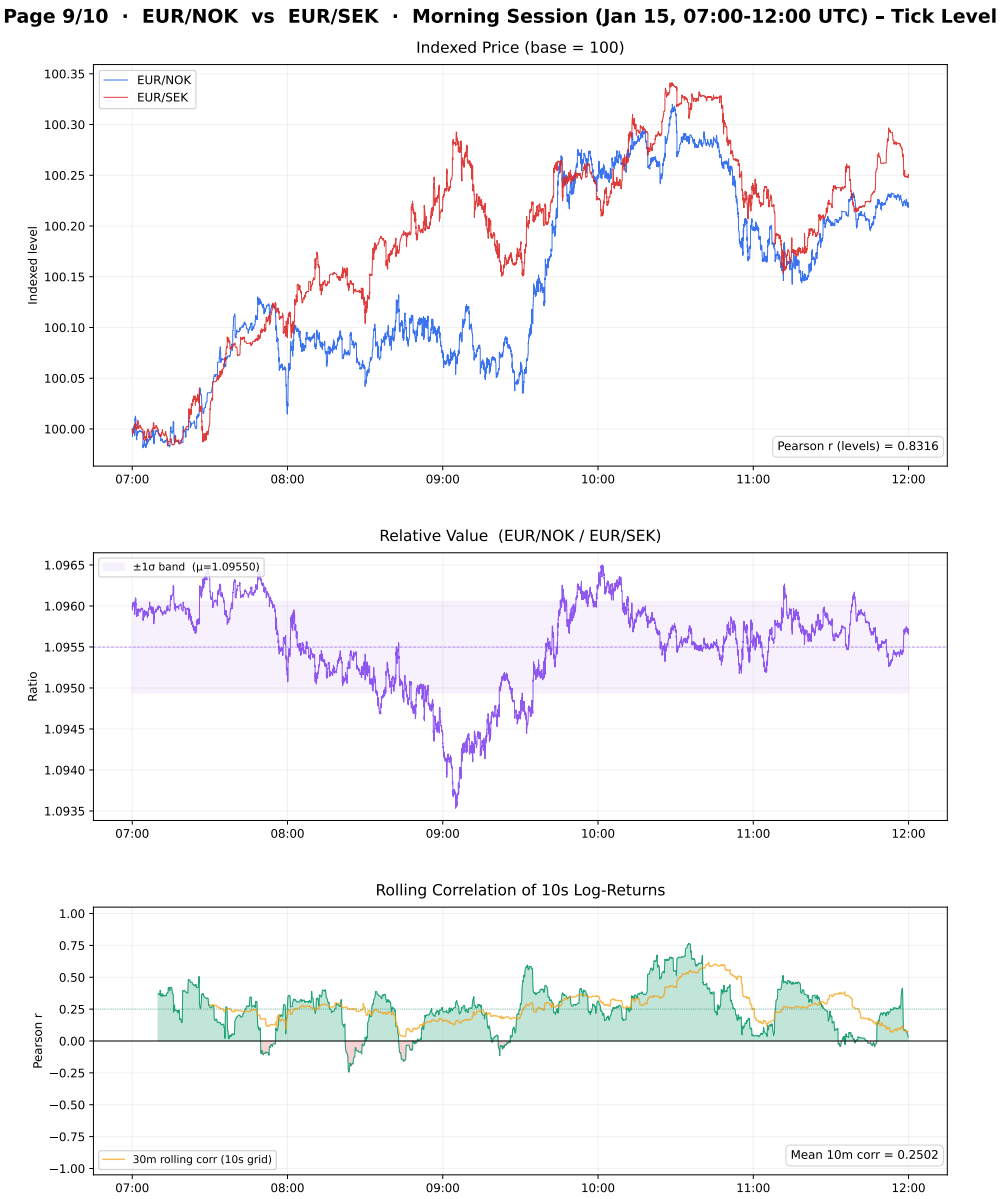


Simulated MS-GARCH process



EUR/DKK exchange rate

Applied Case: Pairs Trading



Thank you